

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2058

Boone Dam 138 kV
October 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2058. The Interconnection Customer proposes a 187.6 MW solar photovoltaic facility interconnecting at the Boone Dam 138 kV substation in the Tennessee area. The request was received on April 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 588 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,398,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BOONE DAM to BAYOU COVE PEAKING POWER 138kV	TELLICO RES DEV AUTH to BOONE DAM 138kV	156.9	204	100.0
TELLICO RES DEV AUTH to BOONE DAM 138kV Line	BAYOU COVE PEAKING POWER to BOONE DAM SUBSTATION 138kV	144.4	222	100.0
LEBANON to BYHALIA 138kV	KEAN MTN AND GRASSY CRK to BARR LAKE SUBSTATION 345kV	141.8	236	6.2
CALPEAK PE 7 SUBSTATION to BYHALIA 138kV	BARR LAKE to KEAN MTN AND GRASSY CRK 345kV L	134.6	240	6.2
CLINCH RIVER to WINNEBAGO WIND POWER PROJECT 138kV	KEAN MTN AND GRASSY CRK to BARR LAKE 345kV	129.6	280	6.2
DAFFIN GIN to CALPEAK PE 7 138kV L	BARR LAKE to KEAN MTN AND GRASSY CRK 345kV	119.3	264	6.2
LOCHMERE HYDROELECTRIC PLANT to HINTON AP 138kV	MANATEE SUBSTATION to HINTON AP 345kV	118.3	265	6.9
PITCHGEN to LOCHMERE HYDROELECTRIC PLANT 138kV	MANATEE to HINTON AP 345kV	117.4	272	6.9
BRIAR to PITCHGEN 138kV L	MANATEE to HINTON AP SUBSTATION 345kV	116.5	419	10.3
WINNEBAGO WIND POWER PROJECT to LEBANON 138kV	KEA MTN AND GRASSY CREEK to BARR LAKE 345kV L	116.3	300	6.2
BRIAR to FRASER 138kV L	HINTON AP to MANATEE 345kV Line	116.2	422	10.3
INGLES ST to CLAYTOR 138kV Line	LOWMOOR to HINTON AP 345kV line	111.1	216	5.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STOCK ISLAND to KENTUCKY DAM 138kV line	KEAN MTN AND GRASSY CREEK to BARR LAKE 345kV Line	108.8	503	7.2
PRINGLE SUBSTATION to HALLSRIDGE 138kV	NEW HOPE AP to MANATEE 345kV	108.5	243	6.8
MCADENVILLE to VACHERIE 138kV	STOCK ISLAND to KENTUCKY DAM 138kV	107.3	417	7.3
LORE 345/138 kV Auto	LORE to SIGEL WIND PARK 345kV line	107.3	452	7.0
S PRINCETON to HALLS RIDGE 138kV	MANATEE to NEW HOPE AP 345kV	107.1	269	6.8
NESPELEM to OPTICAL COATING LAB 138kV L	BARR LAKE to NEW HOPE AP 345kV	106.0	316	7.5
NESPELEM to PIUTE 138kV	BARR LAKE to NEW HOPE AP 345kV	105.8	319	7.5
BAR LAKE to NEW HOPE AP SUBSTATION 345kV	BARR LAKE to PARKE 345kV line	105.5	1,174	25.2
KEA MTN AND GRASSY CRK to CLINCH RIV 345kV	KEAN MTN AND GRASSY CREEK 345-1 kV Auto	102.4	1,254	25.2
CLINCH RIVER SUBSTATION 345/138 kV Transformer	BARR LAKE to KEAN MTN AND GRASSY CREEK 345kV	102.1	355	6.2
OROVADA to CARLTON HILLS 138kV	ATOKA to CARLTON HILLS 345kV	102.0	733	9.5
BARR LAKE to KEAN MTN AND GRASSY CREEK 345kV	STATION 11 to STOCK ISLAND 138kV	101.8	1,483	29.3
DRESDEN to POINT ARENA 500kV	CONESVILLE STATION to DRESDEN 500kV line	101.5	2,973	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BOONE DAM to BAYOU COVE PEAKING POWER 138kV	Rebuild line with higher-capacity conductor	\$4,723,000	55% (\$2,572,000)
TELLICO RES DEV AUTH to BOONE DAM 138kV Line	Rebuild line with higher-capacity conductor	\$4,000,000	55% (\$2,178,000)
LEBANON to BYHALIA 138kV	Rebuild line with higher-capacity conductor	\$8,371,000	7% (\$589,000)
BYHALIA to CALPEAK PE 7 138kV	Rebuild line with higher-capacity conductor	\$19,118,000	7% (\$1,345,000)
CLINCH RIVER to WINNEBAGO WIND POWER PROJECT 138kV	Rebuild line with higher-capacity conductor	\$5,444,000	7% (\$383,000)
DAFFIN GIN to CALPEAK PE 7 138kV L	Rebuild line with higher-capacity conductor	\$3,405,000	7% (\$240,000)
LOCHMERE HYDROELECTRIC PLANT to HINTON AP 138kV	Rebuild line with higher-capacity conductor	\$2,368,000	4% (\$102,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PITCHGEN to LOCHMERE HYDROELECTRIC PLANT 138kV	Rebuild line with higher-capacity conductor	\$4,643,000	4% (\$200,000)
BRIAR to PITCHGEN 138kV line	Rebuild line with higher-capacity conductor	\$5,590,000	4% (\$234,000)
WINNEBAGO WIND POWER PROJECT to LEBANON 138kV	Rebuild line with higher-capacity conductor	\$3,529,000	7% (\$248,000)
BRIAR to FRASER 138kV L	Rebuild line with higher-capacity conductor	\$2,407,000	4% (\$101,000)
INGLES ST to CLAYTOR 138kV Line	Rebuild line with higher-capacity conductor	\$4,495,000	5% (\$235,000)
KENTUCKY DAM SUBSTATION to STOCK ISLAND 138kV L	Reconductor line	\$856,000	3% (\$27,000)
PRINGLE to HALLS RIDGE 138kV	Reconductor line	\$3,946,000	5% (\$209,000)
MCADENVILLE to VACHERIE 138kV	Reconductor line	\$6,573,000	3% (\$169,000)
LORE 345/138 kV Auto	Replace transformer with higher-rated unit	\$14,163,000	4% (\$592,000)
HALLS RIDGE to SOUTH PRINCETON 138kV	Reconductor line	\$1,309,000	5% (\$69,000)
NESPELEM to OPTICAL COATING LAB 138kV L	Reconductor line	\$1,603,000	4% (\$66,000)
NESPELEM to PIUTE 138kV	Reconductor line	\$2,830,000	4% (\$117,000)
NEW HOPE AP to BARR LAKE SUBSTATION 345kV	Reconductor line	\$23,623,000	4% (\$1,005,000)
CLINCH RIV SUBSTATION to KEAN MTN AND GRASSY CREEK SUBSTATION 345kV	Reconductor line	\$26,817,000	4% (\$1,209,000)
CLINCH RIVER 345/138kV XFMR	Replace transformer with higher-rated unit	\$14,880,000	7% (\$1,047,000)
CARLTON HILLS to OROVADA 138kV	Reconductor line	\$221,000	2% (\$5,000)
BARR LAKE to KEAN MTN AND GRASSY CREEK 345kV	Reconductor line	\$22,128,000	4% (\$931,000)
DRESDEN to PT ARENA 500kV Line	Reconductor line	\$68,830,000	4% (\$2,525,000)
	Total allocated to project		\$16,398,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.